Chem 150 Term Test 1B 14 February 2025 8:30 Student Name:_____

1.(**2 marks**) Using the periodic table arrange the following elements in the expected order of increasing ionization energy: Ca, Se, Br, Sr

Sr, Ca, Se, Br ✓

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2.(**3 marks**) Arrange the following ions in order increasing size: **I^-, Ba^{2+}, Sb^{3-}, Sr^{2+}**

I^- Ba^{2+} Sb^{-3} Sr^{2+}

54 54 54 32

Sr^{2+}, Ba^{2+}, I^-, Sb^{-3} ✓

3. (**6 marks**) Consider the following four bonds between elements shown below. Using the electronegativity values provided on the last page of this exam, assign partial (fractional) charges (δ^+, δ^-,) to all eight atoms. Next rank the molecules in order of increasing bond polarity (increasing % ionic character).

B-F	N-O	C-O	Si-H
2.0 - 4.0	3.0-3.5	2.5 - 3.5	1.8 - 2.1
δ^+ δ^-	δ^+ δ^-	δ^+ δ^-	δ^+ δ^-
Si - H	N - O	C - O	B - F

(less) ionic ← → more ionic ✓

4.(**6 marks**) Name three types (extremes) of chemical bonding and give a representative example for each type. Your example must be a pure substance (e.g. an element or a compound)

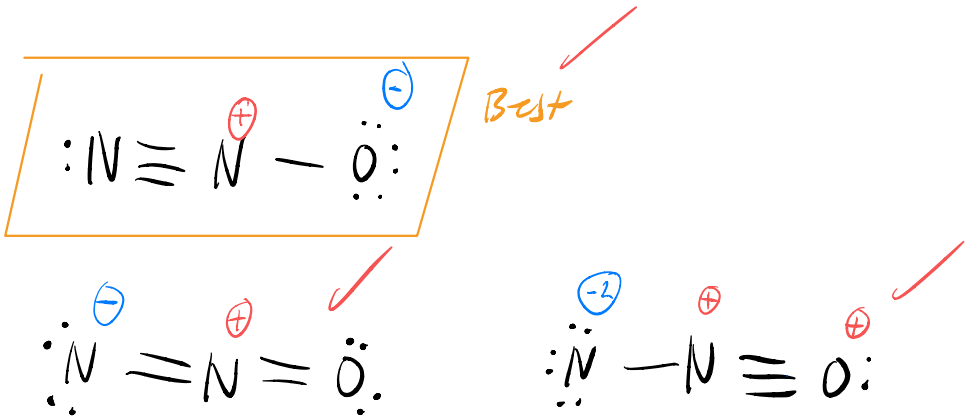
a) Ionic – NaCl – Sodium Chloride – salt

b) Covalent – H_2O – water

c) Metalic – Fe – Iron

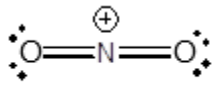
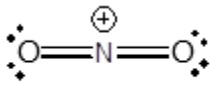
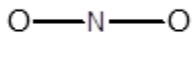
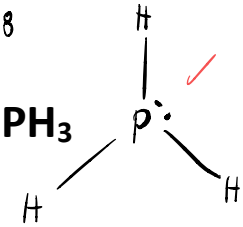
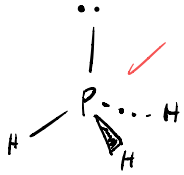
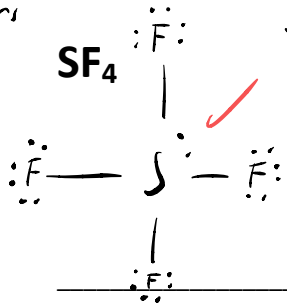
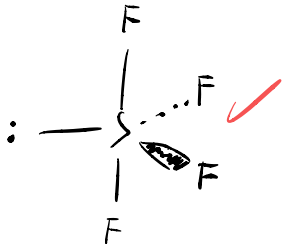
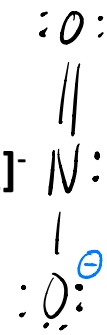
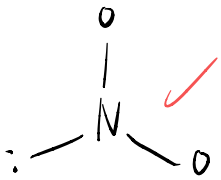
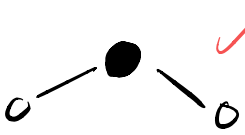
5. (**8 marks**) Draw all valid Lewis structures of the molecule N_2O. If there are formal charges on any of the atoms be sure to include them! Indicate if the structures you have drawn are equivalent or non-equivalent. Circle equivalent Resonance structures. If you have non-equivalent resonance structures, state this also and underline your best resonance structure.

16 VE = 8 pairs



None are equivalent

4.(**21 marks**) Identify the electron group geometry (EGG) and the molecular shape of the species below. Start by writing a valid Lewis structure and state the number of electron groups (#EG) around the central atom. Be sure to show all electrons and indicate any non-zero formal charges. Continue by drawing the EGG placing lone pairs and bonding pairs in the appropriate positions. **Name the EGG**. Next draw the molecular shape. Here multiple bonds, formal charges on atoms and lone pairs might be omitted for clarity. **Name the molecular shape**. Finally, indicate if the molecule or ion is polar or not. Please refer to the example below (SF_4) if in doubt about any of the instructions.

	Lewis Structure	#EG	EGG	shape	polar (Yes or No)
		2	 linear	 linear	No
5 + 3 = 8 4 pairs	**PH_3** 	4	 tetrahedral	tripod	Yes
34 VE = 17 pairs	**SF_4** 	5	 trigonal bipyramidal	Seesaw / canada goose	Yes
18 VE 4 pairs	**$[NO_2]^-$** 	3	 trigonal planar	 Bent 120°	Yes

4

7.(**8 marks**) Using the data given below estimate the lattice energy for magnesium fluoride, MgF_2 in kJ/mol. Note that not all the data given below are necessary to complete this task. Drawing a diagram with energy along the y axis might prove invaluable. Positive energies mean energy is required (moving up along the y-axis) whereas negative energies mean energy is released (going down along the y-axis).

~~Melting Point Magnesium: 650 °C~~

~~Melting point Fluorine: -219 °C~~

~~Boiling point Fluorine: -188 °C~~

Heat of sublimation for magnesium: 146 kJ/mol

~~Density of magnesium: 1.74 g/cm³~~

~~First ionization energy fluorine: 1681 kJ/mol~~

~~2nd Ionization energy of fluorine: 3373 kJ/mol~~

electron affinity of fluorine: -328 kJ/mol

First ionization energy magnesium: 738 kJ/mol

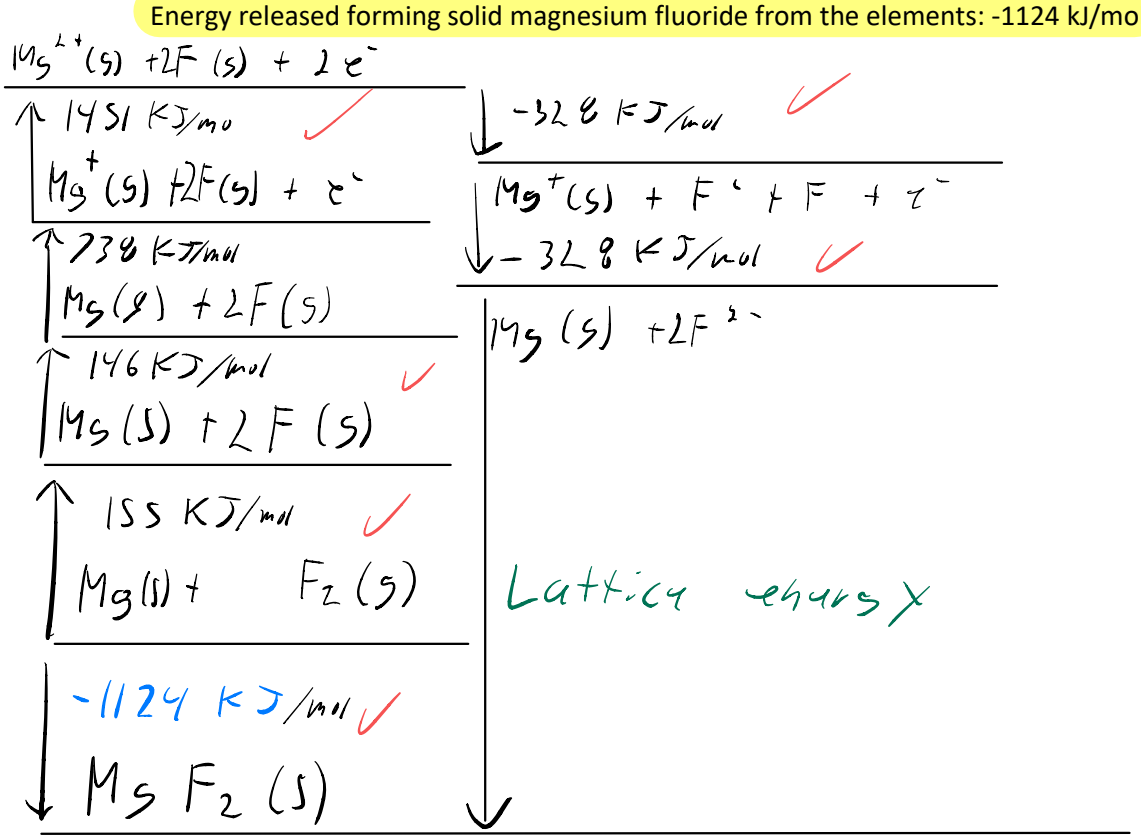
Second ionization energy magnesium: 1451 kJ/mol

~~Third ionization energy magnesium: 2652.5 kJ/mol~~

~~Fourth ionization energy Titanium: 7733 kJ/mol~~

Bond dissociation energy for the difluorine molecule: 155 kJ/mol

Energy released forming solid magnesium fluoride from the elements: -1124 kJ/mol



$= -328 - 328 + 1451 + 738 + 146 + 155 + 1124 =$

$2958\ kJ/mol$

Electronegativities of the elements

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
H 2.1																
Li 1.0	Be 1.5											B 2.0	C 2.5	N 3.0	O 3.5	F 4.0
Na 0.9	Mg 1.2											Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0
K 0.8	Ca 1.0	Sc 1.3	Ti 1.5	V 1.6	Cr 1.6	Mn 1.5	Fe 1.8	Co 1.8	Ni 1.8	Cu 1.9	Zn 1.6	Ga 1.6	Ge 1.8	As 2.0	Se 2.4	Br 2.8
Rb 0.8	Sr 1.0	Y 1.2	Zr 1.4	Nb 1.6	Mo 1.8	Tc 1.9	Ru 2.2	Rh 2.2	Pd 2.2	Ag 1.9	Cd 1.7	In 1.7	Sn 1.8	Sb 1.9	Te 2.1	I 2.5
Cs 0.8	Ba 0.9	La* 1.1	Hf 1.3	Ta 1.5	W 2.4	Re 1.9	Os 2.2	Ir 2.2	Pt 2.2	Au 2.4	Hg 1.9	Tl 1.8	Pb 1.8	Bi 1.9	Po 2.0	At 2.2
Fr 0.7	Ra 0.9	Ac† 1.1														

below 1.0
1.0–1.4
1.5–1.9
2.0–2.4
2.5–2.9
3.0–4.0

*Lanthanides: 1.1–1.3
†Actinides: 1.3–1.5